

Paper 2

Question		Generic Scheme	Illustrative Scheme	Max Mark
1	a			
• ¹	ss	find gradient of AB	• ¹ $m_{AB} = 1$	4
• ²	pd	find perpendicular gradient	• ² $m_{\text{perp}} = -1$ stated or implied by • ⁴	
• ³	pd	find midpoint of AB	• ³ (4,1) stated or implied by • ⁴	
• ⁴	pd	obtain equation	• ⁴ $y - 1 = -1(x - 4)$	

Notes:

- ⁴ is only available as a consequence of using a perpendicular gradient **and** a midpoint.
- The gradient must appear in simplified form at •⁴ stage for •⁴ to be awarded.

Commonly Observed Responses:

Candidate A

$$m_{AB} = -1 \quad \bullet^1 \text{ X}$$

$$m_{\text{perp}} = 1 \quad \bullet^2 \text{ X}$$

$$(4,1) \quad \bullet^3 \text{ ✓}$$

$$y - 1 = 1(x - 4) \Rightarrow y = x - 3 \quad \bullet^4 \text{ X}$$

Leading to part (b)

$$y - x = -3 \quad \bullet^5 \text{ X}$$

$$y + 2x = 6$$

$$(3,0) \quad \bullet^6 \text{ X}$$

•⁷ and •⁸ are not available as $A = T = (3,0)$

Question		Generic Scheme	Illustrative Scheme	Max Mark
1	b			2
• ⁵	ss	know to solve simultaneously	• ⁵ $y + 2x = 6$ $y + x = 5$	
• ⁶	pd	solve correctly for x and y	• ⁶ $x = 1, y = 4$	
Commonly Observed Responses:				
Candidate B				
Part (a) $y - 1 = -1(x - 4)$ • ⁴ ✓ $y = -x + 3$ error				
Part (b) $y + 2x = 6$ and $y + x = 3$ • ⁵ ✓ $x = 3, y = 0$ • ⁶ ✗ correct strategy used, pd mark not available				
1	c			2
• ⁷	ss	know and use $m = \tan \theta$	• ⁷ $\tan \theta = -2$	
• ⁸	pd	calculate angle	• ⁸ 116.6° accept 117^0 or 2.03 radians	
Commonly Observed Responses:				
Candidate C		Candidate D		
$m_{AT} = -\frac{1}{2}$				
base angle = 26.6° • ⁷ ✗		$m_{AT} = 2$ • ⁷ ✗		
$\Rightarrow$ angle = $90 + 26.6 = 116.6^\circ$ • ⁸ ✗		angle = $\tan^{-1}(2) = 63.4^\circ$ • ⁸ ✓		
Candidate E:				
Part (a)		Part (b)		
$m_{AB} = \frac{2-0}{5-3} = \frac{2}{8} = \frac{1}{4}$ • ¹ ✗		$y + 4x - 5 = 0$ • ⁵ ✗ $\Rightarrow$ $y + 2x = -6$ • ⁶ ✗ $y + 2x + 6 = 0$		
$m_{\text{perp}} = -4$ • ² ✓		$\Rightarrow 2x = 1, x = \frac{1}{2}, y = -7$		
Midpoint of AB (4, 1) • ³ ✓		• ⁵ is a strategy mark. The correct strategy is to solve the given equation with the equation from part (a) simultaneously. • ⁵ is not awarded as the given equation has not been used.		
$y - 1 = -4(x - 1)$ • ⁴ ✓		The equation obtained at stage • ⁴ , has been rearranged incorrectly in part (b). The next pd mark, • ⁶ , is therefore not awarded.		
$y + 4x - 5$				

Question		Generic Scheme		Illustrative Scheme		Max Mark
2						4
• ¹	ss	know to and differentiate		• ¹	$4x^3 - 6x^2$	
• ²	ic	find gradient		• ²	8	
• ³	pd	find y -coordinate		• ³	5	
• ⁴	ic	state equation of tangent		• ⁴	$y - 5 = 8(x - 2)$	
Notes:						
1. • ⁴ is only available if an attempt has been made to find the gradient from differentiation and calculating the y-coordinate by substitution into the original equation.						
Commonly Observed Responses:						
Candidate A						
• ¹ ✓ • ² ✓ • ³ ✓						
using $y = mx + c$						
$x = 2, y = 5, m = 8$						
$\Rightarrow 5 = 8 \times 2 + c$						
$\Rightarrow c = -11$ • ⁴ ✓						
$y = 8x - 11$						

Question		Generic Scheme	Illustrative Scheme	Max Mark
3	a			2
• ¹	ic	interpret notation	• ¹ $f(x+3)$ stated or implied by • ²	
• ²	pd	a correct expression	• ² $= (x+3)(x+2)+q$ OR $= (x+3)^2 - (x+3)+q$ or equivalent	
Notes:				
1. Special Case: • ¹ is for substituting $(x+3)$ for x thus, treat $x+3(x+3-1)+q$ as bad form.				
Commonly Observed Responses:				
Candidate A		Candidate B		
$f(g(x)) = x+3(x+3-1)+q$ $= x^2+5x+6+q$		$f(g(x)) = x+3(x+3-1)+q$ $= 4x+6+q$		
• ¹ ✓ • ² ✓ • ³ ✓		• ¹ ✓ • ² ✗		
Candidate C		Candidate D		
$f(g(x)) = x+3(x+3-1)+q$ $= (x+3)^2 - x+3+q$ $x^2+5x+6+q=0$		$f(g(x)) = (x+3)(x+3-1)+q$ $= (x+3)^2 - x+3+q$ $x^2+5x+12+q=0$		
• ¹ ✓ • ² ✓ • ³ ✓		• ¹ ✓ • ² ✓ • ³ ✗		
Candidate E: using $g(f(x))$				
part (a)		part (b)		
$g(f(x)) = g(x(x-1)+q)$ $= x(x-1)+q+3$		$x^2-x+q+3=0$ $b^2-4ac = (-1)^2 - 4 \times 1 \times (q+3)$ $1-4q-12=0$ $q = -\frac{11}{4}$		
• ¹ ✗ • ² ✗		• ³ ✗ (eased) • ⁴ ✗ • ⁵ ✗ • ⁶ ✗		
Leading to				

Question		Generic Scheme	Illustrative Scheme	Max Mark
3	b			
		Method 1 •³ pd write in standard quadratic form •⁴ ic use discriminant •⁵ pd simplify and equate to zero •⁶ pd find value of q Method 2 •³ pd write in standard quadratic form •⁴ ic complete the square •⁵ pd equate to zero •⁶ pd find value of q Method 3 •³ pd write in standard quadratic form •⁴ ic geometric interpretation •⁵ pd differentiates to obtain x •⁶ pd find value of q	Method 1 •³ $x^2 + 5x + 6 + q = 0$ •⁴ $b^2 - 4ac = 5^2 - 4 \times 1 \times (6 + q)$ •⁵ $\Rightarrow 25 - 24 - 4q = 0$ •⁶ $q = \frac{1}{4}$ Method 2 •³ $x^2 + 5x + 6 + q = 0$ •⁴ $\left(x + \frac{5}{2}\right)^2 - \frac{25}{4} + 6 + q = 0$ •⁵ $-\frac{25}{4} + 6 + q = 0$ •⁶ $q = \frac{1}{4}$ Method 3 •³ $f(g(x)) = x^2 + 5x + 6 + q = 0$ •⁴ equal roots so touches x-axis at SP •⁵ $\Rightarrow \frac{dy}{dx} = 2x + 5 = 0$ $x = -\frac{5}{2}$ •⁶ $\frac{25}{4} - \frac{25}{2} + 6 + q = 0$ $q = \frac{1}{4}$	4

Notes:

- Do not penalise the omission of ' $= 0$ ' at •³.
- In Method 1 $a=1$, $b=5$, $c=6+q$ is sufficient for •³.
- Candidates who assume ' $= 0$ ' and follow through to a correct value of q , •⁶ is still available. In Methods 1 and 2 ' $= 0$ ' must appear at •⁴ or •⁵ for •⁵ to be awarded.
- If the expression obtained at •³ is not a quadratic then •³, •⁴, •⁵ and •⁶ are not available.

Question	Generic Scheme		Illustrative Scheme	Max Mark
Throughout this question treat coordinates written as components, and vice versa, as bad form.				
4	a			2
• ¹	pd	states coordinates of C	• ¹ C(11,12,6)	
• ²	pd	states coordinates of D	• ² D(8,8,4)	
Notes:				
1. Accept $x=11$, $y=12$ and $z=6$ for • ¹ and $x=8$, $y=8$ and $z=4$ for • ² . 2. For candidates who write the coordinates as Cartesian triples and omit brackets in both cases, • ² is not available.				
4	b			2
• ³	pd	finds $\overrightarrow{CB}$	• ³ $\begin{pmatrix} 0 \\ -8 \\ -4 \end{pmatrix}$	
• ⁴	pd	finds $\overrightarrow{CD}$	• ⁴ $\begin{pmatrix} -3 \\ -4 \\ -2 \end{pmatrix}$	
Notes:				
3. For candidates who find both $\overrightarrow{BC}$ and $\overrightarrow{DC}$, only • ⁴ is available (repeated error).				
4	c	.		5
• ⁵	ss	know to use scalar product applied to the correct angle	• ⁵ $\cos \hat{BCD} = \frac{\overrightarrow{CB} \cdot \overrightarrow{CD}}{ \overrightarrow{CB} \overrightarrow{CD} }$ stated or implied by • ⁹	
• ⁶	pd	find scalar product	• ⁶ 40	
• ⁷	pd	find $ \overrightarrow{CB} $	• ⁷ $\sqrt{80}$	
• ⁸	pd	find $ \overrightarrow{CD} $	• ⁸ $\sqrt{29}$	
• ⁹	pd	find angle	• ⁹ 33.9°	
Notes:				
4. • ⁵ is not available for candidates who choose to evaluate an incorrect angle. 5. • ⁹ accept 33.8 to 34 degrees or 0.59 to 0.6 radians. 6. If candidates do not attempt • ⁹ , then • ⁵ is only available if the formula quoted relates to the labelling in the question. 7. • ⁹ is only available as a result of using a valid strategy. 8. • ⁵ is not available for candidates who write $\cos \theta = \frac{40}{\sqrt{80} \times \sqrt{29}}$. Some reference to the labelling of the diagram must be made within their solution to part (c), to indicate they are attempting to find the correct angle.				

Commonly Observed Responses:	
Candidate A: Cosine Rule $\cos \hat{C}D = \frac{CB^2 + CD^2 - BD^2}{2 \times CB \times CD} \quad \bullet^5 \checkmark$ $CB = \sqrt{80}, CD = \sqrt{29}, BD = \sqrt{29} \quad \bullet^6 \checkmark \bullet^7 \checkmark \bullet^8 \checkmark$ $33.9^\circ \text{ or } 0.59 \text{ radians} \quad \bullet^9 \checkmark$	Candidate B $\cos \hat{C}D = \frac{\overline{BC} \cdot \overline{CD}}{ \overline{BC} \times \overline{CD} } \quad \bullet^5 \times$ $\overline{BC} \cdot \overline{CD} = -40 \quad \bullet^6 \times$ $ \overline{BC} = \sqrt{80}, \overline{CD} = \sqrt{29} \quad \bullet^7 \times \bullet^8 \times$ $146.1^\circ \text{ or } 2.55 \text{ radians} \quad \bullet^9 \times$
Candidate C $\cos \hat{B}D = \frac{\overline{OB} \cdot \overline{OD}}{ \overline{OB} \times \overline{OD} } \quad \bullet^5 \times$ $\overline{OB} \cdot \overline{OD} = 128 \quad \bullet^6 \times$ $ \overline{OB} = \sqrt{141}, \overline{OD} = 12 \quad \bullet^7 \times \bullet^8 \times$ $26.1^\circ \text{ or } 0.46 \text{ radians} \quad \bullet^9 \times$	Candidate D $\cos \hat{C}D = \frac{\overline{BC} \cdot \overline{BD}}{ \overline{BC} \times \overline{BD} } \quad \bullet^5 \times$ $\overline{BC} \cdot \overline{BD} = 40 \quad \bullet^6 \times$ $ \overline{BC} = \sqrt{80}, \overline{BD} = \sqrt{29} \quad \bullet^7 \times \bullet^8 \times$ $33.9^\circ \text{ or } 0.59 \text{ radians} \quad \bullet^9 \times$
Candidate E $\cos \hat{B}C = \frac{\overline{OB} \cdot \overline{OC}}{ \overline{OB} \times \overline{OC} } \quad \bullet^5 \times$ $\overline{OB} \cdot \overline{OC} = 181 \quad \bullet^6 \times$ $ \overline{OB} = \sqrt{141}, \overline{OC} = \sqrt{301} \quad \bullet^7 \times \bullet^8 \times$ $28.5^\circ \text{ or } 0.50 \text{ radians} \quad \bullet^9 \times$	Candidate F $\cos \hat{C}D = \frac{\overline{BC} \cdot \overline{DC}}{ \overline{BC} \times \overline{DC} } \quad \bullet^5 \checkmark$ this is an acceptable form for the scalar product.

Question		Generic Scheme	Illustrative Scheme	Max Mark
5				
• ¹	ss	start to integrate	• ¹ $\frac{1}{\frac{1}{2}}(\dots)^{\frac{1}{2}}$	
• ²	pd	complete integration	• ² $\dots \times \frac{1}{3}$	
• ³	pd	process limits	• ³ $\frac{2}{3}(3t+4)^{\frac{1}{2}} - \frac{2}{3}(3(4)+4)^{\frac{1}{2}}$	
• ⁴	pd	start to solve equation	• ⁴ $(3t+4)^{\frac{1}{2}} = 7$	
• ⁵	pd	solve for t	• ⁵ $t = 15$	
				5

Notes:

- ³ is awarded for correct substitution leading to $F(t) - F(4)$ where $F(x)$ is the candidates attempt
- to integrate $(3x+4)^{-\frac{1}{2}}$. For substituting into the original function •³ is unavailable.
- ⁵ is only available as a consequence of squaring both sides of an equation.
- The integral obtained must contain a non integer power for •⁴ and •⁵ to be available.
- Do not penalise the inclusion of '+c'.
- Incorrect expansion of $(\dots)^{-\frac{1}{2}}$ at stage •¹, only •³ is available as follow through. Incorrect expansion of $(\dots)^{\frac{1}{2}}$ at stage •⁴, •⁴ and •⁵ are not available.

Commonly Observed Responses:

Candidate A: Forgetting the $\frac{1}{3}$ $\left[2(3x+4)^{\frac{1}{2}}\right]_4^t = 2$ • ¹ ✓ • ² ✗ $\left(2(3t+4)^{\frac{1}{2}}\right) - \left(2(3(4)+4)^{\frac{1}{2}}\right) = 2$ • ³ ✗ $(3t+4)^{\frac{1}{2}} = 5$ • ⁴ ✗ $t = 7$ • ⁵ ✗	Candidate B $\left[\frac{1}{6}(3x+4)^{\frac{1}{2}}\right]_4^t = 2$ • ¹ ✗ • ² ✗ $\left(\frac{1}{6}(3t+4)^{\frac{1}{2}}\right) - \left(\frac{1}{6}(3(4)+4)^{\frac{1}{2}}\right) = 2$ • ³ ✗ $(3t+4)^{\frac{1}{2}} = 16$ • ⁴ ✗ $t = 84$ • ⁵ ✗
Candidate C $\left[\frac{(3x+4)^{\frac{1}{2}}}{\frac{1}{2}} \times 3\right]_4^t = 2$ • ¹ ✓ • ² ✗ $\left[\frac{2}{3}(3x+4)^{\frac{1}{2}}\right]_4^t = 2$ $\left[\frac{2}{3}(3t+4)^{\frac{1}{2}}\right] - \left[\frac{2}{3}(3(4)+4)^{\frac{1}{2}}\right] = 2$ • ³ ✗ $(3t+4)^{\frac{1}{2}} = 7$ • ⁴ ✗ $t = 15$ • ⁵ ✗	Candidate D $\left[-\frac{3}{2}(3x+4)^{-\frac{3}{2}}\right]_4^t = 2$ • ¹ ✗ • ² ✗ $-\frac{3}{2}(3t+4)^{-\frac{3}{2}} - \left(-\frac{3}{2} \times 16^{-\frac{3}{2}}\right) = 2$ • ³ ✗ $(3t+4)^{\frac{3}{2}} = -\frac{192}{253}$ • ⁴ ✗ decimal equivalent not accepted $t = -1.056$ • ⁵ ✗

Question		Generic Scheme		Illustrative Scheme	Max Mark
6					
	•¹ ss use correct double angle formula •² ss arrange in standard quadratic form •³ ss start to solve •⁴ ic reduce to equations in sin x only •⁵ pd process to find solutions in given domain	•¹ $\sin x - 2(1 - 2\sin^2 x)$ stated or implied by •² •² $4\sin^2 x + \sin x - 3 = 0$ •³ $(4\sin x - 3)(\sin x + 1) = 0$ OR $\frac{-1 \pm \sqrt{(1)^2 - 4 \times 4 \times (-3)}}{2 \times 4}$ •⁴ $\sin x = \frac{3}{4}$ and $\sin x = -1$ •⁵ $0.848, 2.29$ and $\frac{3\pi}{2}$ OR •⁴ $\sin x = \frac{3}{4}$ and $x = 0.848, 2.29$ •⁵ $\sin x = -1$, and $x = \frac{3\pi}{2}$			5

Notes:

- ¹ is not available for simply stating $\cos 2A = 1 - 2\sin^2 A$ with no further working.
- In the event of $\cos^2 x - \sin^2 x$ or $2\cos^2 x - 1$ being substituted for $\cos 2x$, •¹ cannot be awarded until the equation reduces to a quadratic in $\sin x$.
- Substituting $1 - 2\sin^2 A$ or $1 - 2\sin^2 \alpha$ for $\cos 2\alpha$ at •¹ stage should be treated as bad form provided the equation is written in terms of x at stage •². Otherwise, •¹ is not available.
- '=0' must appear by •³ stage for •² to be awarded. However, for candidates using the quadratic formula to solve the equation, '=0' must appear at •² stage for •² to be awarded.
- Candidates may express the equation obtained at •² in the form $4s^2 + s - 3 = 0$ or $4x^2 + x - 3 = 0$. In these cases, award •³ for $(4s - 3)(s + 1) = 0$ or $(4x - 3)(x + 1) = 0$. However, •⁴ is only available if $\sin x$ appears explicitly at this stage.
- ⁴ and •⁵ are only available as a consequence of solving a quadratic equation.
- ³, •⁴ and •⁵ are not available for any attempt to solve a quadratic written in the form $ax^2 + bx = c$.
- ⁵ is not available to candidates who work in degrees and do not convert their solutions into radian measure.
- $\sin x + 4\sin^2 x - 3 = 0$ does not gain •², unless •³ is awarded.

Commonly Observed Responses:			
Candidate A		Candidate B	
$\bullet^1 \checkmark \bullet^2 \checkmark$ $(4s-3)(s+1)=0$ $s=\frac{3}{4}, s=-1$ $x=0.848, 2.29 \text{ and } \frac{3\pi}{2}$	$\bullet^3 \checkmark$ $\bullet^4 \times$ $\bullet^5 \checkmark$	$\bullet^1 \checkmark$ $4\sin^2 x + \sin x - 3 = 0$ $5\sin x - 3 = 0$ $\sin x = \frac{3}{5}$ $x = 0.644, 2.50$	$\bullet^2 \checkmark$ $\bullet^3 \times$ $\bullet^4 \times$ $\bullet^5 \times$
Candidate C		Candidate D	
$\bullet^1 \checkmark$ $\sin x - 2(1 - 2\sin^2 x) = 1$ $\sin x - 2 + 4\sin^2 x = 1$ $4\sin^2 x + \sin x = 3$ $\sin x(4\sin x + 1) = 3$ $\sin x = 3, 4\sin x + 1 = 3$ no solution, $\sin x = \frac{1}{2}$ $x = \frac{\pi}{6}, \frac{5\pi}{6}$	$\bullet^2 \times$ $\bullet^3 \times$ $\bullet^4 \times$ $\bullet^5 \times$	$\bullet^1 \checkmark$ $\sin x - 2(1 - 2\sin^2 x) = 1$ $4\sin^2 x + \sin x - 3 = 0$ $4\sin^2 x + \sin x = 3$ $\sin x(4\sin x + 1) = 3$ $\sin x = 3, 4\sin x + 1 = 3$ no solution, $\sin x = \frac{1}{2}$ $x = \frac{\pi}{6}, \frac{5\pi}{6}$	$\bullet^2 \checkmark$ $\bullet^3 \times$ $\bullet^4 \times$ $\bullet^5 \times$
Candidate E: Reading $\cos 2x$ as $\cos^2 x$ $\sin x - 2\cos^2 x = 1$ $\bullet^1 \times$ $\sin x - 2(1 - \sin^2 x) = 1$ $2\sin^2 x + \sin x - 3 = 0$ $\bullet^2 \times$ $(2\sin x + 3)(\sin x - 1) = 0$ $\bullet^3 \times$ $\sin x = -\frac{3}{2}, \sin x = 1$ $\bullet^4 \times$ no solution, $x = \frac{\pi}{2}$ $\bullet^5 \times$			

Question		Generic Scheme		Illustrative Scheme	Max Mark
7	a				5
• ¹	ss	know to and find intersection of line and curve	• ¹	$2x = 6x - x^2 \Rightarrow x = 0, x = 4$	
• ²	ic	use “ upper – lower”	• ²	$\int \left((6x - x^2) - 2x \right) dx$	
• ³	pd	integrate	• ³	$2x^2 - \frac{1}{3}x^3$	
• ⁴	pd	substitute limits and evaluate	• ⁴	$10\frac{2}{3}$	
• ⁵	pd	evaluate area developed	• ⁵	$10\frac{2}{3} \times 300 = 3200 \text{ m}^2$	
Notes:					
1. ‘0’ appearing as the lower limit of the integral is sufficient evidence for $x = 0$ at •¹ stage. 2. •⁵ is only available as a consequence of multiplying an exact answer at •⁴ stage. 3. The omission of dx at •² should not be penalised. 4. Where a candidate differentiates one or both terms •³, •⁴ and •⁵ are unavailable. 5. Do not penalise the inclusion of ‘+ c’. 6. Accept $\int (4x - x^2) dx$ for •².					

Commonly Observed Responses:

Candidate A

$$\int_0^4 (2x - (6x - x^2)) dx \quad \bullet^2 \text{ X}$$

$$= \frac{1}{3}x^3 - 2x^2 \quad \bullet^3 \text{ X}$$

$$= -10\frac{2}{3} \text{ cannot be negative so } = 10\frac{2}{3} \quad \bullet^4 \text{ X} \quad \text{however } \dots = -10\frac{2}{3} \text{ so Area } = 10\frac{2}{3} \quad \bullet^4 \text{ X}$$

$$\text{Area} = 3200\text{m}^2 \quad \bullet^5 \text{ X}$$

Candidate B

$$2x = 6x - x^2 \Rightarrow x = 0, 4 \quad \bullet^1 \text{ X}$$

Shaded area

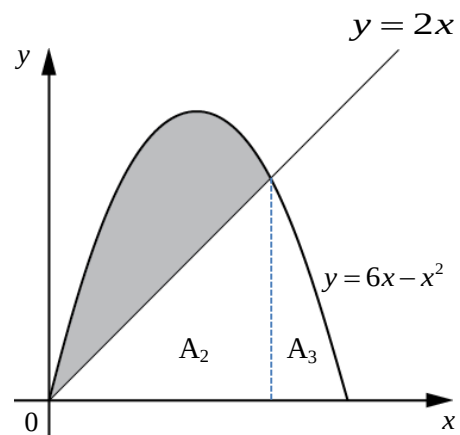
$$= \text{area under parabola} - (A_2 + A_3)$$

$$= \int_0^6 (6x - x^2) dx - \left[A_2 + \int_4^6 (6x - x^2) dx \right] \quad \bullet^2 \text{ X}$$

Stated or implied by $\bullet^4$

$$\text{Area under parabola} = 36, A_2 = 16 \text{ and } A_3 = \frac{28}{3} \quad \bullet^3 \text{ X}$$

$$\text{Shaded area} = 36 - \left(16 + \frac{28}{3} \right) = \frac{32}{3} \quad \bullet^4 \text{ X}$$



Candidate C

Part (a)

$$x = 0, x = 6 \quad \bullet^1 \text{ X}$$

$$\int ((6x - x^2) - 2x) dx \quad \bullet^2 \text{ X}$$

$$\left[2x^2 - \frac{1}{3}x^3 \right]_0^6 \quad \bullet^3 \text{ X}$$

$$\left(2 \times 6^2 - \frac{1}{3} \times 6^3 \right) - (0) = 0 \quad \bullet^4 \text{ X}$$

$$\Rightarrow \text{Area} = 0 \times 300 = 0 \text{ m}^2 \quad \bullet^5 \text{ X}$$

Question		Generic Scheme	Illustrative Scheme	Max Mark
7	b			
• ⁶	ss	set derivative to 2	• ⁶ $6 - 2x = 2$	5
• ⁷	pd	find point of contact	• ⁷ $x = 2, y = 8$	
• ⁸	pd	find equation of road	• ⁸ $y = 2x + 4$	
• ⁹	ss	find correct integral	• ⁹ $\left[(x^2 + 4x) - \left(3x^2 - \frac{1}{3}x^3 \right) \right]_0^2$	
• ¹⁰	ic	calculate area	• ¹⁰ 800m^2	

Notes:

- For candidates who omit 'm²' at both •⁵ and •¹⁰ stages, •¹⁰ is not available.
- Candidates who arrive at an incorrect equation at •⁸, or produce an equation ex nihilo, must use an equation of the form $y = 2x + c$ with $c > 0$, for •⁹ and •¹⁰ to be available.
- $y = 2x + 4$ must appear explicitly or as part of the integrand for •⁸ to be awarded.
- ¹⁰ is only available as a result of a valid strategy at the •⁹ stage,
ie $\int (\text{line}) - (\text{quadratic})$ **and** lower limit = 0 and upper limit < 3.

Commonly Observed Responses:

Candidate D: Alternative Method

Line has equation of the form $y = 2x + c$, $y = 2x + c$ and $y = 6x - x^2$

intersect where $x^2 - 4x + c = 0$

•⁶ ✓

tangency $\Rightarrow$ 1 point of intersection

$$\Rightarrow b^2 - 4ac = 0$$

•⁷ ✓

$$16 - 4c = 0$$

•⁸ ✓

$$c = 4$$

Continue as above.

Question		Generic Scheme	Illustrative Scheme	Max Mark
8				
• ¹	pd	correct values	• ¹ $g = -p, f = -2p, c = 3p + 2$	5
• ²	ss	substitute and rearrange	• ² $5p^2 - 3p - 2$	
• ³	ic	knowing condition	• ³ $g^2 + f^2 - c > 0$	
• ⁴	pd	factorise and solve	• ⁴ $(5p + 2)(p - 1) = 0 \Rightarrow p = -\frac{2}{5}, p = 1$	
• ⁵	ic	correct range	• ⁵ $p < -\frac{2}{5}, p > 1$	

Notes:

- Candidates who state the coordinates of the centre, $(p, 2p)$ and state the radius, $r = \sqrt{\dots - (3p + 2)}$ gain •¹.
- Accept $(-p)^2 + (-2p)^2 - (3p + 2)$ or $p^2 + (2p)^2 - (3p + 2)$. If brackets are omitted •¹ may only be awarded if subsequent working is correct.
- Do not accept $(-p)^2 + (2p)^2 - (3p + 2)$ or $(p)^2 + (-2p)^2 - (3p + 2)$ for •¹.
- Do not accept $g^2 + f^2 - c \geq 0$ for •³.
- For a candidate who uses $c = 2$ and follows through to get $p < -\sqrt{\frac{2}{5}}, p > \sqrt{\frac{2}{5}}$, award •², •³ and •⁵.
- Evidence for •³ may appear at •⁵ stage.
- ⁴ and •⁵ can only be awarded for solving a quadratic inequation.

Commonly Observed Responses:

Candidate A		Candidate B	
$g = -2p, f = -4p, c = 3p + 2$	• ¹ ✗	$(x - p)^2 - p^2 + (y - 2p)^2 - 4p^2 + 3p + 2 = 0$	
$20p^2 - 3p - 2$	• ² ✓	$(x - p)^2 + (y - 2p)^2$	• ¹ ✓
$g^2 + f^2 - c > 0$	• ³ ✓	$= 5p^2 - 3p - 2$	• ² ✓
$(4p + 1)(5p - 2) = 0 \Rightarrow p = -\frac{1}{4}, p = \frac{2}{5}$	• ⁴ ✓	$5p^2 - 3p - 2 > 0$	• ³ ✓
$p < -\frac{1}{4}, p > \frac{2}{5}$	• ⁵ ✓	$(5p + 2)(p - 1) > 0$	• ⁴ ✓
		$p < -\frac{2}{5}, p > 1$	• ⁵ ✓

Question		Generic Scheme	Illustrative Scheme	Max Mark
9	a			
• ¹	ss	know to differentiate	• ¹ $a = v'(t)$	3
• ²	pd	differentiates trig. function	• ² $-8\sin\left(2t - \frac{\pi}{2}\right) \dots\dots$	
• ³	pd	applies chain rule	• ³ $\dots\dots \times 2$ and complete $a(t) = -16\sin\left(2t - \frac{\pi}{2}\right)$	

Commonly Observed Responses:

Candidate A: Alternative Method

Part (a)

$$v(t) = 8\cos\left(2t - \frac{\pi}{2}\right) = 8\sin 2t$$

$$v'(t) = \dots \quad \bullet^1 \checkmark$$

$$= 8\cos 2t \dots$$

$$\bullet^2 \checkmark$$

$$= \dots\dots \times 2 \quad \bullet^3 \checkmark$$

Part (b)

$$v'(10) = 16\cos 20 = 6.53$$

$$\bullet^4 \checkmark$$

$> 0, \Rightarrow$ velocity is increasing $\bullet^5 \checkmark$

Part (c)

$$s(t) = \int v(t) dt \quad \bullet^6 \checkmark$$

$$s(t) = -4\cos 2t + c \quad \bullet^7 \checkmark$$

$$4 = -4 + c \Rightarrow c = 8$$

$$\Rightarrow s(t) = -4\cos 2t + 8 \quad \bullet^8 \checkmark$$

$$\text{or } \Rightarrow s(t) = 8 - 4\cos 2t$$

Candidate B: Candidates who misinterpret the process for rate of change.

Part (a)

$$a(t) = \int 8\cos\left(2t - \frac{\pi}{2}\right) dt$$

$$= 4\sin\left(2t - \frac{\pi}{2}\right) + c$$

Wrong process award $\frac{0}{3}$

Part (b)

$$\text{If } t = 10, a = 4\sin\left(20 - \frac{\pi}{2}\right) + c$$

$$= -1.63 + c$$

Cannot evaluate award $\frac{0}{2}$

Part (c)

$$s = v'(t)$$

$$s(t) = -16\sin\left(2t - \frac{\pi}{2}\right)$$

Award $\frac{2}{3}$

Candidate C

Part (a)

$$a = v'(t) \text{ or equivalent} \quad \bullet^1$$

$$a = 4\sin\left(2t - \frac{\pi}{2}\right) \quad \bullet^2 \times \quad \bullet^3 \times$$

Part (b)

$$a(10) = 4\sin\left(20 - \frac{\pi}{2}\right) = -1.63 \quad \bullet^4$$

< 0 , So decreasing $\bullet^5 \checkmark$

Only as a consequence of $\bullet^1$ in part (a)

Question		Generic Scheme		Illustrative Scheme		Max Mark
9	b					2
• ⁴	ss	know to and evaluate $a(10)$		• ⁴	$a(10) = 6.53$	
• ⁵	ic	interpret result		• ⁵	$a(10) > 0$ therefore increasing	
Notes:						
1. • ⁵ is available only as a consequence of substituting into a derivative.						
2. • ⁴ and • ⁵ are not available to candidates who work in degrees.						
3. • ² and • ³ may be awarded if they appear in the working for 9(b). However, • ¹ requires a clear link between acceleration and $v'(t)$.						
9	c					3
• ⁶	ic	know to integrate		• ⁶	$s(t) = \int v(t) dt$	
• ⁷	pd	integrate correctly		• ⁷	$s(t) = 4 \sin \left(2t - \frac{\pi}{2} \right) + c$	
• ⁸	ic	determine constant and complete		• ⁸	$c = 8$ so $s(t) = 4 \sin \left(2t - \frac{\pi}{2} \right) + 8$	
Notes:						
4. • ⁷ and • ⁸ are not available to candidates who work in degrees. However, accept $\int 8 \cos (2t - 90) dt$ for • ⁶ .						

[END OF MARKING INSTRUCTIONS]